

Probabilistic Methods

Lecture Notes

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Chapter 1

Introduction

These notes introduce probabilistic methods as a way of reasoning about uncertainty. The course begins with sample spaces and events, then develops counting techniques, probability distributions, standard distribution functions, conditional probability, and finally descriptive statistics based on generated data.

The central aim is to connect formal probability with interpretation. A probability calculation should not be treated as a mechanical substitution into a formula. It should begin with a clear description of the random experiment, the elementary outcomes, the events or random variables of interest, and the assumptions that make a chosen model appropriate.

Chapter 2

Events and Probability (Sample Spaces)

This chapter introduces the fundamental concepts of probability theory, including sample spaces, events, and the calculation of probabilities.

Chapter 3

Combinatorics for Probability

This chapter covers the combinatorial methods that are essential for calculating probabilities in a systematic way.

Chapter 4

Probability Distributions (Selected Models)

This chapter introduces several important probability distributions that are used to model various random phenomena.

Chapter 5

Discovering Formalism

This chapter focuses on the conceptual development of probability theory, moving from intuitive ideas to a more formal mathematical framework.

Chapter 6

PMF, PDF, and CDF

This chapter provides a technical overview of the standard representations of probability distributions: the Probability Mass Function (PMF), the Probability Density Function (PDF), and the Cumulative Distribution Function (CDF).

Chapter 7

Algebra of Events and Conditional Probability

This chapter explores the algebra of events and the concept of conditional probability, which are fundamental to understanding how probabilities are updated with new information.

Chapter 8

Descriptive Statistics from Generated Data

This chapter shifts the perspective from theoretical probability models to the analysis of data using descriptive statistics.